QUESTION_ID: 55ea0659

QUESTION:

$x(r - 9) + 4 = 19x + 27$

In the given equation, r is a positive integer. If the given equation has exactly one solution, what CANNOT be the value of r?

ANSWER_CHOICES:

A. 4

B. 9

C. 23

D. 28

END_QUESTION

QUESTION_ID: 088bc84e

QUESTION:

If $2(5x) = 6$, what is the value of $5x$?

ANSWER_CHOICES:

A. 4/5

B. 5/3

C. 3

D. 4

END_QUESTION

QUESTION_ID: f8fa0cf0

QUESTION:

$9r = 7(r + 7)$

What value of r is the solution to the given equation?

ANSWER_CHOICES:

[FREE_RESPONSE]

END_QUESTION

QUESTION_ID: 5eb4918d

QUESTION:

If $9x + 4 = 67$, what is the value of $90x + 40$?

ANSWER_CHOICES:

A. 7

B. 70

C. 130

D. 670

END_QUESTION

QUESTION_ID: 935f3403

QUESTION:

For a restaurant owner to purchase a pizza oven at a total price of 13,000 dollars, a onetime down payment is required, and then fixed monthly payments are made for the remaining amount owed for the pizza oven. The equation $13{,}000 = 2{,}800 + 200t$ represents this situation, where t is the number of fixed monthly payments that are made. Which of the following is the best interpretation of 200 in this context?

ANSWER_CHOICES:

A. The amount, in dollars, of the down payment

B. The amount, in dollars, of each fixed monthly payment

C. The total amount, in dollars, paid for the pizza oven after t fixed monthly payments

D. The total number of fixed monthly payments

END_QUESTION

QUESTION_ID: 30b1e186

QUESTION:

In the xy-plane, which of the following does NOT contain any points that are part of the solution set to $5x - 7y > 35$?

ANSWER_CHOICES:

A. The x-axis

B. The region where $x > 0$ and $y > 0$

C. The region where $x < 0$ and $y < 0$

D. The region where $x < 0$ and $y > 0$

END_QUESTION

QUESTION_ID: d914abec

QUESTION:

To complete a landscaping project, Adam charges a fee of $20.00 for his equipment and $9.50 per hour spent working on the project. To complete this same landscaping project, Caroline charges a fee of $17.00 for her equipment and $10.00 per hour spent working on the project. If x represents the number of hours spent working on the landscaping project, what are all the values of x for which Caroline's total charge is greater than Adam's total charge?

ANSWER_CHOICES:

A. $5 \le x \le 6$

B. $6 \le x \le 7$

C. $x < 5$

D. $x > 6$

END_QUESTION

QUESTION_ID: e8fd6d99

QUESTION:

Terry has a goal to collect at least 16 items per week for a donation drive. This week, Terry has collected 4 items. If x represents the additional number of items Terry needs to collect this week to meet the goal, which inequality represents this situation?

ANSWER_CHOICES:

A. $4 - x \leq 16$

B. $4 + x \leq 16$

C. $4 - x \geq 16$

D. $4 + x \geq 16$

END_QUESTION

QUESTION_ID: 59813abf

QUESTION:

$38(x - n) = 38y + 38n$

One of the equations in a system of two linear equations is given, where n is a positive constant. The system has no solution. Which equation could be the second equation in this system?

ANSWER_CHOICES:

A. $4x - 4y = 8n$

B. $4x + 4y = 4n$

C. $4x + 4y = 8n$

D. $4x - 4y = 4n$

END_QUESTION

QUESTION_ID: ea8bd66f

QUESTION:

$y = 4x$

$5x - 3y = -14$

The solution to the given system of equations is (x, y). What is the value of $x + y$?

ANSWER_CHOICES:

A. 4

B. 6

C. 8

D. 10

END_QUESTION

QUESTION_ID: 5e9b6079

QUESTION:

$y = 7x + 18$

One of the equations in a system of two linear equations is given. The system has no solution. Which equation could be the second equation in the system?

ANSWER_CHOICES:

A. $-21x + y = 54$

B. $-21x + y = 18$

C. $-7x + y = 21$

D. $-7x + y = 18$

END_QUESTION

QUESTION_ID: 18ac8354

QUESTION:

The graph of line g is shown in the xy-plane. The line passes through (-10, 12) and (0, 1). Line k is defined by 165x + py = w, where p and w are constants. If line k is graphed in this xy-plane, resulting in the graph of a system of two linear equations, the system of two linear equations will have infinitely many solutions. What is the value of p + w?

ANSWER_CHOICES:

[FREE_RESPONSE]

END_QUESTION

QUESTION_ID: 0e926898

QUESTION:

Students and chaperones from a middle school are going on a field trip to a museum. Admission to the museum will cost $8.50 for each student and $12 for each chaperone. It will cost a total of $730 for admission to the museum for all 83 people on the field trip. How many students are going on the field trip?

ANSWER_CHOICES:

A. 31

B. 70

C. 76

D. 86

END_QUESTION

QUESTION_ID: 340afb12

QUESTION:

Ellen bought two types of snack bags for a school event. Each box of chips Ellen bought contained 24 bags and cost $11. Each box of crackers Ellen bought contained 16 bags and cost $9. It cost Ellen $497 before tax to buy 968 bags. If x is the number of boxes of chips and y is the number of boxes of crackers that Ellen bought, which of the following systems of equations represents this situation?

ANSWER_CHOICES:

A. $11x + 16y = 968$; $24x + 9y = 497$

B. $11x + 9y = 968$; $24x + 16y = 497$

C. $11x + 9y = 497$; $24x + 16y = 968$

D. $9x + 11y = 497$; $16x + 24y = 968$

END_QUESTION

QUESTION_ID: 080e75c2

QUESTION:

The functions f and g model the number of participants, in thousands, in two different programs x years since 2010. The graphs of y = f(x) and y = g(x) are shown. The decreasing line starts at (0, 13), the increasing line starts at (0, 7), and the lines intersect at approximately (11, 11). Which of the following could represent functions f and g?

ANSWER_CHOICES:

A. f(x) = -(4/11)x + 13; g(x) = (2/11)x + 7

B. f(x) = -(4/11)x + 7; g(x) = (2/11)x + 13

C. f(x) = -(2/11)x + 13; g(x) = (4/11)x + 7

D. f(x) = -(2/11)x + 7; g(x) = (4/11)x + 13

END_QUESTION

QUESTION_ID: d4572f55

QUESTION:

For a linear relationship between x and y, the table gives three values of x and their corresponding values of y, where t is a constant.

TABLE: $x = -34$, $y = t$; $x = -17$, $y = t + 23$; $x = 0$, $y = t + 46$.

Which equation represents this relationship?

ANSWER_CHOICES:

A. $y = -2x + t + 23$

B. $y = 2x + t + 23$

C. $y = -(23/17)x + t + 46$

D. $y = (23/17)x + t + 46$

END_QUESTION

QUESTION_ID: 1a2a0f59

QUESTION:

A yoga studio is offering a deal where the first session is free, the second session is half off the regular price, and the remaining sessions are at the regular price. If the regular price of a session is $25.40, which function f gives the total cost, in dollars, of purchasing x sessions using this deal, where $x \geq 2$?

ANSWER_CHOICES:

A. $f(x) = 25.40(x - 1) + 12.70$

B. $f(x) = 25.40(x - 2) + 12.70$

C. $f(x) = 25.40(x - 1) + 12.70(x - 2)$

D. $f(x) = 25.40(x - 2) + 12.70(x - 1)$

END_QUESTION

QUESTION_ID: 590d662d

QUESTION:

While walking on a trail, Mary stopped to read a trail sign. The graph shows the total distance y, in meters, Mary had walked on the trail x minutes after leaving the trail sign. The line passes through (0, 150) and (10, 650). What distance, in meters, did Mary walk on the trail in the 10 minutes after leaving the trail sign?

ANSWER_CHOICES:

[FREE_RESPONSE]

END_QUESTION

QUESTION_ID: fc4a0d2a

QUESTION:

$g(x) = 3(14x - 15)$

What is the y-coordinate of the y-intercept of the graph of $y = g(x) - 2$ in the xy-plane?

ANSWER_CHOICES:

A. -47

B. -45

C. -17

D. -15

END_QUESTION

QUESTION_ID: 2493c767

QUESTION:

$s(x) = 133 - 4x$

The function s shown gives the number of spoons, $s(x)$, remaining in a dispenser x minutes after the dispenser had been loaded with spoons. Which statement is the best interpretation of $s(3) = 121$?

ANSWER_CHOICES:

A. The number of spoons remaining in the dispenser decreased by a total of 3 spoons after 121 minutes.

B. There were 3 spoons remaining in the dispenser 121 minutes after the dispenser had been loaded with spoons.

C. There were 121 spoons remaining in the dispenser 3 minutes after the dispenser had been loaded with spoons.

D. The number of spoons remaining in the dispenser decreased by a total of 121 spoons after 3 minutes.

END_QUESTION

QUESTION_ID: 0c490cd5

QUESTION:

Sam has a flag with an area of 110 square inches (in^2). Sam plans to enlarge the flag by sewing on pieces of colored nylon, where each piece would increase the flag's area by 64 in^2. Which equation gives the flag's total area y, in square inches, after Sam sews on x pieces of nylon?

ANSWER_CHOICES:

A. y = 64x

B. y = 64 + 110x

C. y = 110 + 64x

D. y = 174x

END_QUESTION

QUESTION_ID: 107d0c62

QUESTION:

$0.10x + 0.20y = 0.17(x + y)$

The equation gives a volume x, in gallons, of a 10% solution that could be mixed with a volume y, in gallons, of a 20% solution to produce a 17% solution. According to this equation, what volume, in gallons, of the 20% solution could be mixed with 90.0 gallons of the 10% solution to produce a 17% solution? (Assume that the volume of the mixture is the sum of the volumes of the two solutions before they were mixed.)

ANSWER_CHOICES:

[FREE_RESPONSE]

END_QUESTION

QUESTION_ID: 4722abd2

QUESTION:

Line p is defined by $2y + 8x = 11$. Line r is perpendicular to line p in the xy-plane. What is the slope of line r?

ANSWER_CHOICES:

A. -4

B. -1/4

C. 1/4

D. 4

END_QUESTION

QUESTION_ID: f6c18a66

QUESTION:

For data set A, the table summarizes the distribution of the number of deliveries received by an office each day during a period of 11 days.

TABLE: Deliveries 0 -> 2 days; 3 -> 2 days; 4 -> 2 days; 5 -> 2 days; 6 -> 1 day; 7 -> 1 day; 13 -> 1 day.

The data value 13 was recorded in error and is removed from data set A to create data set B, which consists of the remaining 10 data values. Which statement best compares the median of data set A and the median of data set B?

ANSWER_CHOICES:

A. The median of data set B is less than the median of data set A.

B. The median of data set B is greater than the median of data set A.

C. The median of data set B is equal to the median of data set A.

D. There is not enough information to compare the medians of the two data sets.

END_QUESTION

QUESTION_ID: d73a67d7

QUESTION:

Which of the following lists represents a data set with the smallest standard deviation?

ANSWER_CHOICES:

A. 51, 52, 54, 56, 57

B. 52, 52, 54, 56, 56

C. 52, 53, 54, 55, 56

D. 53, 54, 54, 54, 55

END_QUESTION

QUESTION_ID: 89ff6a0a

QUESTION:

For 100 buttons, the table summarizes the distribution by group and diameter.

TABLE: Columns = Less than 20 mm, 20 to 30 mm, Greater than 30 mm. Group 1 = 12, 8, 3. Group 2 = 0, 17, 18. Group 3 = 10, 32, 0.

One of these buttons will be selected at random. What is the probability of selecting a button with a diameter that is less than or equal to 30 millimeters, given that it is not in group 2? (Express your answer as a decimal or fraction, not as a percent.)

ANSWER_CHOICES:

[FREE_RESPONSE]

END_QUESTION

QUESTION_ID: 8bb94668

QUESTION:

The speeds of objects A, B, and C are a meters per second, b meters per second, and c meters per second, respectively. If the speed of object A is 5,900% of the speed of object C and the speed of object C is 0.008% of the speed of object B, which expression represents the value of a + b in terms of c?

ANSWER_CHOICES:

A. 1,840c

B. 5,908c

C. 6,025c

D. 12,559c

END_QUESTION

QUESTION_ID: dbd89c59

QUESTION:

The scatterplot shows the relationship between the number of swans, y, in a certain population and the number of years since 1970, x, where $0 \leq x \leq 50$. The plotted data are approximately (0, 43), (10, 65), (20, 85), (30, 150), (40, 190), and (50, 290). Which of the following equations is the most appropriate exponential model for the data shown in the scatterplot?

ANSWER_CHOICES:

A. $y = 43(1.04)^{-x}$

B. $y = 43(1.04)^{x}$

C. $y = 288(1.04)^{-x}$

D. $y = 288(1.04)^{x}$

END_QUESTION

QUESTION_ID: b3547a10

QUESTION:

A group of 10 botanists recorded data on the germination rates of a certain type of seed for one growing season. The scatterplot shows the relationship between the number of seeds planted, x, and the number of seeds that germinated, y, for each of the botanists. A line of best fit is also shown. Which of the following is the best interpretation of the slope of the line of best fit in this context?

ANSWER_CHOICES:

A. The number of seeds planted is predicted to increase by 60 seeds every 100 days.

B. The number of seeds planted is predicted to increase by 300 seeds every 100 days.

C. The number of seeds that germinate is predicted to increase by 60 seeds for every additional 100 seeds that are planted.

D. The number of seeds that germinate is predicted to increase by 300 seeds for every additional 100 seeds that are planted.

END_QUESTION

QUESTION_ID: 4c5ea142

QUESTION:

On average, a certain plant grows 87 millimeters every m months. At this rate, which expression represents the number of millimeters, on average, the plant grows every k years?

ANSWER_CHOICES:

A. 29m/(4k)

B. 29k/(4m)

C. 1,044m/k

D. 1,044k/m

END_QUESTION

QUESTION_ID: 1b96c116

QUESTION:

A rectangle has an area of 3,168 square inches. What is the area, in square feet, of this rectangle? (1 foot = 12 inches)

ANSWER_CHOICES:

A. 22

B. 56.3

C. 264

D. 675.4

END_QUESTION

QUESTION_ID: 2bab1399

QUESTION:

A researcher is designing a study to investigate the average number of hours participants in a training course spend on a particular activity per day. The researcher will report an estimated average number of hours participants in the training course spend on this activity per day with an associated margin of error. The researcher is considering using a random sample of either 115 or 230 participants from the training course. Which of the following would be the most likely effect of using the larger random sample compared to the smaller random sample?

ANSWER_CHOICES:

A. The reported margin of error would be lower.

B. The reported margin of error would be higher.

C. The reported average number of hours would be lower.

D. The reported average number of hours would be higher.

END_QUESTION

QUESTION_ID: f496d2c0

QUESTION:

The table shows the distribution of rooms in a certain facility by seating capacity.

TABLE: Less than 18 seats = 26%; 18-40 seats = 21%; 41-65 seats = 29%; Greater than 65 seats = 24%.

If a room in this facility is selected at random, which of the following is closest to the probability of selecting a room that has a seating capacity greater than 65 seats, given that the room has a seating capacity of at least 18 seats?

ANSWER_CHOICES:

A. 0.24

B. 0.32

C. 0.50

D. 0.92

END_QUESTION

QUESTION_ID: da978ee6

QUESTION:

A researcher surveyed a random sample of 1,020 people from a certain city to estimate the percentage of people who support a recently announced proposal. From the survey, the researcher estimated that 86% of people from the city support this proposal, with an associated margin of error of 2.13%. The researcher repeated the survey with a random sample of people from the city that was double the original sample size. Assuming the margins of error were calculated in the same way, which of the following is true about the margin of error associated with the estimate from the larger sample?

ANSWER_CHOICES:

A. The margin of error is between 2.13% and 3.13%.

B. The margin of error is between 4.13% and 5.13%.

C. The margin of error is greater than 5.13%.

D. The margin of error is less than 2.13%.

END_QUESTION

QUESTION_ID: 6feae9c3

QUESTION:

At a recreation center, visitors used the exercise room for a total of t hours in July. At the same recreation center, visitors used the exercise room for $3.49t$ hours in August. What is the percent increase in the number of hours visitors used the exercise room from July to August?

ANSWER_CHOICES:

A. 2.49%

B. 3.49%

C. 249%

D. 349%

END_QUESTION

QUESTION_ID: 559476f4

QUESTION:

The bar graph shows the frequency of each data value in a data set. The frequencies are: data value 1 -> 4, 2 -> 6, 3 -> 3, 4 -> 8, 5 -> 5, 6 -> 9. What is the frequency of data value 2 in this data set?

ANSWER_CHOICES:

[FREE_RESPONSE]

END_QUESTION

QUESTION_ID: 70a2776f

QUESTION:

As part of a study of lake conditions, the water temperature was recorded at different depths below the surface of a certain lake. The scatterplot gives the recorded temperature, in degrees Celsius, for 7 depths, in meters, below the surface of the lake. Which of the following is closest to the slope of a line of best fit for the data shown?

ANSWER_CHOICES:

A. -0.17

B. -2.17

C. -6.02

D. -8.02

END_QUESTION

QUESTION_ID: de799680

QUESTION:

A cluster fly is traveling at a velocity of 76 centimeters per second. What is this velocity, in millimeters per second? (1 centimeter = 10 millimeters)

ANSWER_CHOICES:

A. 7,600

B. 836

C. 760

D. 86

END_QUESTION

QUESTION_ID: 98818acf

QUESTION:

The number of audiobooks in a library increased by 131% from last year to this year. The number of audiobooks in the library last year was n, and the number of audiobooks in the library this year is bn, where b and n are constants. What is the value of b?

ANSWER_CHOICES:

[FREE_RESPONSE]

END_QUESTION

QUESTION_ID: f01108a8

QUESTION:

$y = 9(a/7)^{(x + c)} - b$

How many times does the graph of the given equation in the xy-plane cross the x-axis, where a, b, and c are positive constants such that $a > 7$ and $b > c$?

ANSWER_CHOICES:

A. Zero

B. One

C. Two

D. Three

END_QUESTION

QUESTION_ID: 71b74a44

QUESTION:

$y = 7{,}400(0.87)^x$

The given equation estimates the value, in dollars, of a certain piece of equipment x years after it was purchased. According to the given equation, what was the estimated value, in dollars, of the piece of equipment at the time it was purchased?

ANSWER_CHOICES:

A. 740

B. 6,438

C. 7,400

D. 8,700

END_QUESTION

QUESTION_ID: 3cf2698e

QUESTION:

The number of cells of a certain bacteria doubles every 25 minutes during its growth period when placed in a medium of milk in a controlled laboratory environment. If a sample of 830 cells of this bacteria is placed in milk in a controlled laboratory environment, which function f best represents the number of cells of the bacteria x minutes after the growth period begins, where x is less than the number of minutes in the growth period?

ANSWER_CHOICES:

A. $f(x) = 830(1/2)^{(x/25)}$

B. $f(x) = 830(2)^{(x/25)}$

C. $f(x) = 830(1/3)^{(x/25)}$

D. $f(x) = 830(3)^{(x/25)}$

END_QUESTION

QUESTION_ID: 58b4c6f3

QUESTION:

The function h is defined by $h(x) = a^x + b$, where a and b are positive constants. The graph of $y = h(x)$ in the xy-plane passes through the points (0, 10) and (2, 13). What is the value of ab?

ANSWER_CHOICES:

A. 13

B. 18

C. 20

D. 26

END_QUESTION

QUESTION_ID: 7bbfbe70

QUESTION:

A lab analyst observes a sample of a substance. An exponential model estimates that the mass, in grams, of the sample decreases by 24% every 22.96 minutes. Which of the following equations could represent this model, where M is the estimated mass, in grams, of the sample t minutes after the lab analyst began observing the sample?

ANSWER_CHOICES:

A. $M = 100(0.24)^{(t + 22.96)}$

B. $M = 100(0.24)^{(t/22.96)}$

C. $M = 100(0.76)^{(t + 22.96)}$

D. $M = 100(0.76)^{(t/22.96)}$

END_QUESTION

QUESTION_ID: a1e65979

QUESTION:

The function f is a quadratic function. In the xy-plane, the graph of $y = f(x)$ has a vertex at (1, 9) and passes through the points (2, 33) and (-1, 105). What is the value of $f(-2) - f(0)$?

ANSWER_CHOICES:

A. 81

B. 129

C. 192

D. 225

END_QUESTION

QUESTION_ID: 5acbdc30

QUESTION:

The function f is defined by $f(x) = 56(0.19)^x$. For any positive integer n, the value of $f(n)$ is p% less than the value of $f(n - 1)$. What is the value of p?

ANSWER_CHOICES:

A. 19

B. 44

C. 56

D. 81

END_QUESTION

QUESTION_ID: c95686e9

QUESTION:

$h(t) = -16t^2 + b$

The function h estimates a ball's height, in feet, above a floor t seconds after the ball is released, where b is a constant. The function estimates that the ball is 19.36 feet above the floor when it is released at t = 0. How many seconds after being released does the function estimate the ball will reach the floor?

ANSWER_CHOICES:

[FREE_RESPONSE]

END_QUESTION

QUESTION_ID: fe62f031

QUESTION:

The function f is defined by $f(x) = 3x^5$. In the xy-plane, the graph of $y = g(x)$ is the result of shifting the graph of $y = f(x)$ up 4 units. Which equation defines the function g?

ANSWER_CHOICES:

A. $g(x) = 3x^5 + 4$

B. $g(x) = 3x^5 - 4$

C. $g(x) = 3x^{20}$

D. $g(x) = 12x^5$

END_QUESTION

QUESTION_ID: 46308566

QUESTION:

$r^2 + qr = 2r - 55$

In the given equation, q is an integer constant. The given equation has no real solutions. What is the largest possible value of q?

ANSWER_CHOICES:

[FREE_RESPONSE]

END_QUESTION

QUESTION_ID: e833de9a

QUESTION:

$x^2 + y^2 = 36$

$y = mx + b/4$

In the given system of equations, m and b are negative constants. In the xy-plane, the graphs of the equations in the given system intersect at the point (-5, y), where y < 0. Which expression represents the value of b?

ANSWER_CHOICES:

A. $-5m/4 + \sqrt{11}/4$

B. $5m/4 - \sqrt{11}/4$

C. $-20m + 4\sqrt{11}$

D. $20m - 4\sqrt{11}$

END_QUESTION

QUESTION_ID: 5da5c665

QUESTION:

$4x^2 - 5x - 5 = 0$

What is the greatest solution to the given equation?

ANSWER_CHOICES:

A. $5/8 - \sqrt{105}/8$

B. $5/8 + \sqrt{105}/8$

C. $5/4 - \sqrt{105}/4$

D. $5/4 + \sqrt{105}/4$

END_QUESTION

QUESTION_ID: 6bd40794

QUESTION:

$(x - 16)(x - 10)(x + 7)(x + 17) = 0$

What is a positive solution to the given equation?

ANSWER_CHOICES:

[FREE_RESPONSE]

END_QUESTION

QUESTION_ID: c8db0e19

QUESTION:

$1/(4xy) + xyz = 1/(3yz)$

In the given equation, x, y, and z are positive numbers. Which expression is equivalent to y?

ANSWER_CHOICES:

A. $(4x - 3z)/(12x^2z^2)$

B. $\sqrt{(4x - 3z)/(12x^2z^2)}$

C. $1/(3xz^2 - 4x^2z)$

D. $\sqrt{1/(3xz^2 - 4x^2z)}$

END_QUESTION

QUESTION_ID: 5f10c095

QUESTION:

The graph of a system of a linear equation and a quadratic equation is shown. The quadratic is a downward-opening parabola with vertex approximately (5, 8), and the line is horizontal at approximately y = 7.2. A solution to the system is (x, y). What is a possible value of x?

ANSWER_CHOICES:

A. 4

B. 5

C. 7.2

D. 8

END_QUESTION

QUESTION_ID: f123e039

QUESTION:

The graph of a system of a linear and a quadratic equation is shown. The line passes through the origin with negative slope, and the parabola opens upward with vertex (0, -8). Which system of equations is represented by the graph?

ANSWER_CHOICES:

A. $y = -5x$; $y = x^2 + 8$

B. $y = -5x$; $y = x^2 - 8$

C. $y = 5x$; $y = x^2 + 8$

D. $y = 5x$; $y = x^2 - 8$

END_QUESTION

QUESTION_ID: 9a1f1941

QUESTION:

$|4x - 3| = -9$

How many solutions does the given equation have?

ANSWER_CHOICES:

A. Zero

B. One

C. Two

D. Infinitely many

END_QUESTION

QUESTION_ID: 30596346

QUESTION:

$(1/7)(x - k)(16 - x^2) = 0$

In the given equation, k is a positive constant. The sum of the solutions to the equation is 59. What is the value of k?

ANSWER_CHOICES:

[FREE_RESPONSE]

END_QUESTION

QUESTION_ID: 38c90632

QUESTION:

The expression $(29x + 102)/(x(x + 51))$ is equivalent to $p/x + w/(x + 51)$, where p and w are constants. What is the value of w?

ANSWER_CHOICES:

[FREE_RESPONSE]

END_QUESTION

QUESTION_ID: 00165291

QUESTION:

The expression $6x^4 + 31x^2 + 35$ can be rewritten as $(3x^2 + a)(2x^2 + b)$, where a and b are positive integers, or as $(3x^2 + c)(2x^2 + d)$, where c and d are positive nonintegers. What is the value of $a + c$?

ANSWER_CHOICES:

[FREE_RESPONSE]

END_QUESTION

QUESTION_ID: 04887c7b

QUESTION:

$f(x) = 2x + 3$

$g(x) = 7x - 2$

$h(x) = 5x + 6$

The functions f, g, and h are defined as shown. If $f(x) \cdot g(x) - h(x) = ax^2 + bx + c$, where a, b, and c are constants, what is the value of b?

ANSWER_CHOICES:

A. -5

B. 12

C. 20

D. 22

END_QUESTION

QUESTION_ID: 91aa9aa9

QUESTION:

Which expression is equivalent to $(9x^2 + 6) - (8x^2)$?

ANSWER_CHOICES:

A. 7

B. $17x^2 + 6$

C. $x^2 - 6$

D. $x^2 + 6$

END_QUESTION

QUESTION_ID: f8871ccd

QUESTION:

Which expression is equivalent to $2(x - 9/2)(x + 8)$?

ANSWER_CHOICES:

A. $2x^2 - 72$

B. $2x^2 + 7x - 72$

C. $2x^2 - x - 36$

D. $x^2 + 7x - 36$

END_QUESTION

QUESTION_ID: efc469c4

QUESTION:

$-14(5x - 2)^2 + 6(5x - 3)^2$

The given expression can be rewritten as $(a/6)x^2 + (b/6)x + c/6$, where a, b, and c are constants. What is the value of $a + b + c$?

ANSWER_CHOICES:

A. -612

B. -306

C. -102

D. -17

END_QUESTION

QUESTION_ID: 2289b199

QUESTION:

For all positive values of n and p, consider the expression $\sqrt[3]{n^{17} p^5} / [5n^2 \cdot \sqrt[8]{p^8}]$. The given expression is equivalent to which of the following?

I. $[n^{(14/3)} \cdot p^{(5/3)}] / (5np)$

II. $\sqrt[3]{n^{11} p^2} / 5$

ANSWER_CHOICES:

A. I only

B. II only

C. I and II

D. Neither I nor II

END_QUESTION

QUESTION_ID: 1fe41c6b

QUESTION:

The table gives the areas and perimeters of two similar rectangles, where n is a constant.

TABLE: Rectangle A: area = 630 square inches, perimeter = 210 inches. Rectangle B: area = 2,520 square inches, perimeter = n inches.

What is the value of n?

ANSWER_CHOICES:

A. 2,100

B. 1,680

C. 840

D. 420

END_QUESTION

QUESTION_ID: 124bc42b

QUESTION:

Rectangle ABCD shown has a length of 7 units and a width of 2 units. What is the area, in square units, of rectangle ABCD?

ANSWER_CHOICES:

A. 5

B. 9

C. 14

D. 28

END_QUESTION

QUESTION_ID: 30aa51b7

QUESTION:

A rectangle is formed by the four points shown in the xy-plane. The plotted points are (-4, -1), (-1, -5), (4, 5), and (7, 1). What is the area, in square units, of the rectangle?

ANSWER_CHOICES:

[FREE_RESPONSE]

END_QUESTION

QUESTION_ID: 179edb12

QUESTION:

A right circular cone has a height of 22 centimeters (cm) and a base with a diameter of 18 cm. The volume of this cone is $n\pi$ cm^3. What is the value of n?

ANSWER_CHOICES:

[FREE_RESPONSE]

END_QUESTION

QUESTION_ID: c8ab85ad

QUESTION:

Which expression represents the area, in square inches, of a square with a side length of 35 inches?

ANSWER_CHOICES:

A. $\sqrt{35}$

B. 35 + 35

C. (4)(35)

D. 35^2

END_QUESTION

QUESTION_ID: e6c557f9

QUESTION:

The edge length, in inches, of cube Y is 3/88 the edge length, in inches, of cube X. The surface area, in square inches, of cube Y is n times the surface area, in square inches, of cube X. What is the value of n?

ANSWER_CHOICES:

A. 9/7,744

B. 27/3,872

C. 3/88

D. 9/44

END_QUESTION

QUESTION_ID: f3f06c3a

QUESTION:

In the rectangle shown, $x = 7$ centimeters and $y = 5$ centimeters. What is the area, in square centimeters, of the rectangle?

ANSWER_CHOICES:

A. 7

B. 12

C. 35

D. 40

END_QUESTION

QUESTION_ID: 73e1b793

QUESTION:

Right rectangular prism P is similar to right rectangular prism Q. The volume of prism P is 972 cubic centimeters (cm^3), and the volume of prism Q is 36 cm^3. One edge of prism P has a length of 12 cm. What is the length, in cm, of the corresponding edge of prism Q?

ANSWER_CHOICES:

[FREE_RESPONSE]

END_QUESTION

QUESTION_ID: cab8f907

QUESTION:

The figure shows a right triangle with perpendicular side lengths 9 units and 4 units. What is the area, in square units, of the triangle?

ANSWER_CHOICES:

A. 18

B. 13

C. 9

D. 4

END_QUESTION

QUESTION_ID: 9fdc4eaa

QUESTION:

A right square pyramid has a total surface area of 70,560 square inches, and the combined surface area of the four lateral faces of this pyramid is 38,160 square inches. What is the height, in inches, of this pyramid?

ANSWER_CHOICES:

A. 56

B. 90

C. 106

D. 180

END_QUESTION

QUESTION_ID: f6ca90cc

QUESTION:

A right prism has a square base with side lengths of 10 inches. The height of the prism is 58 inches. What is the volume, in cubic inches, of the prism?

ANSWER_CHOICES:

[FREE_RESPONSE]

END_QUESTION

QUESTION_ID: 25cc5327

QUESTION:

The base of a right rectangular pyramid has a length of 27 centimeters and a width of 2 centimeters. The height of this pyramid is 7 centimeters. What is the volume, in cubic centimeters, of this pyramid?

ANSWER_CHOICES:

[FREE_RESPONSE]

END_QUESTION

QUESTION_ID: 4757123b

QUESTION:

Trapezoid A and trapezoid B are similar. The length of each side of trapezoid A is 41 times the length of the corresponding side of trapezoid B. The area of trapezoid A is how many times as large as the area of trapezoid B?

ANSWER_CHOICES:

[FREE_RESPONSE]

END_QUESTION

QUESTION_ID: 17049054

QUESTION:

The length of a side of square X is 9 centimeters. The area of rectangle Y is 32 square centimeters. What is the total area, in square centimeters, of square X and rectangle Y?

ANSWER_CHOICES:

A. 145

B. 113

C. 82

D. 81

END_QUESTION

QUESTION_ID: 42155bac

QUESTION:

The area of one of the bases of a right circular cylinder is 81 square centimeters, and the volume of the cylinder is 2,187 cubic centimeters. What is the height, in centimeters, of the cylinder?

ANSWER_CHOICES:

A. 27

B. 2,025

C. 2,106

D. 177,147

END_QUESTION

QUESTION_ID: 1b687ae3

QUESTION:

The area of a triangle is 36 square units. If the height of this triangle is 2 units, what is the length, in units, of the base of this triangle?

ANSWER_CHOICES:

[FREE_RESPONSE]

END_QUESTION

QUESTION_ID: d12bca0a

QUESTION:

The circle shown has its center at (0, 0). The point (8, 0) lies on the circle, and the point (2, k) lies on the circle. What is the value of k?

ANSWER_CHOICES:

A. 6

B. 7

C. 8

D. $\sqrt{60}$

END_QUESTION

QUESTION_ID: 408b39ea

QUESTION:

$(x - 3)^2 + (y + 16)^2 = 289$

The graph of the given equation is a circle in the xy-plane. The point (q, s) lies on the circle. Which of the following is NOT a possible value of q?

ANSWER_CHOICES:

A. -17

B. -3

C. 3

D. 17

END_QUESTION

QUESTION_ID: 8493fd10

QUESTION:

$x^2 + x\sqrt{t} + y^2 - y - 65 = 0$

The given equation, where t is a constant, defines a circle in the xy-plane. The radius of this circle is $\sqrt{83}$. What is the value of t?

ANSWER_CHOICES:

A. 73

B. 71

C. 37

D. 35

END_QUESTION

QUESTION_ID: 267f82b5

QUESTION:

A circle in the xy-plane has the equation $(x - 19)^2 + (y - 17)^2 = 64k^2$, where k is a positive constant. Which of the following gives the center of the circle and its radius?

ANSWER_CHOICES:

A. The center is at (19, 17), and the radius is 8.

B. The center is at (19, 17), and the radius is 8k.

C. The center is at (19, 17), and the radius is 64.

D. The center is at (19, 17), and the radius is 64k.

END_QUESTION

QUESTION_ID: a5ff1a96

QUESTION:

In the xy-plane, circle M is the graph of the equation $(x - 4)^2 + (y - 5)^2 = 9$. Circle P has the same center as circle M but has a radius that is twice the radius of circle M. Which equation represents circle P?

ANSWER_CHOICES:

A. $(x - 4)^2 + (y - 5)^2 = 18$

B. $(x - 4)^2 + (y - 5)^2 = 36$

C. $(x - 8)^2 + (y - 10)^2 = 9$

D. $(x - 8)^2 + (y - 10)^2 = 18$

END_QUESTION

QUESTION_ID: 186fdbca

QUESTION:

A circle has a diameter of 12.6 centimeters. What is the radius of the circle, in centimeters?

ANSWER_CHOICES:

[FREE_RESPONSE]

END_QUESTION

QUESTION_ID: 149021da

QUESTION:

A circle has center P, and points A and B lie on the circle. The measure of arc AB is $45°$ and the length of arc AB is 4π units. What is the length, in units, of the radius of the circle?

ANSWER_CHOICES:

[FREE_RESPONSE]

END_QUESTION

QUESTION_ID: 11301fb6

QUESTION:

The difference between the measure of angle A and the measure of angle B is $-(5/6)\pi$ radians. Which expression shows the difference between the measure of angle A and the measure of angle B, in degrees?

ANSWER_CHOICES:

A. $-(5/6)\pi \cdot (360°/\pi)$

B. $-(5/6)\pi \cdot (180°/\pi)$

C. $-(5/6)\pi \cdot (\pi/180°)$

D. $-(5/6) \cdot (180°/\pi)$

END_QUESTION

QUESTION_ID: 10e059a6

QUESTION:

The measure of angle A is $\pi/216$ radians. The measure of angle B is 36 times the measure of angle A. What is the value of cos B?

ANSWER_CHOICES:

A. 0

B. 1/2

C. $\sqrt{2}/2$

D. $\sqrt{3}/2$

END_QUESTION

QUESTION_ID: 27f1db42

QUESTION:

In the xy-plane, circle F is defined by the equation $(x + 8)^2 + (y + 8)^2 = 25$. Circle G has the same center as circle F, and the radius of circle G is 2 units greater than the radius of circle F. Circle G is defined by the equation $(x + 8)^2 + (y + 8)^2 = p$, where p is a constant. What is the value of p?

ANSWER_CHOICES:

[FREE_RESPONSE]

END_QUESTION

QUESTION_ID: 44d67c6c

QUESTION:

$x^2 + y^2 - 10x - 8y - 14n = 0$

The given equation represents circle A in the xy-plane, where n is a constant. Point (11, 8) lies on circle B, which has the same center but twice the diameter as circle A. What is the value of n?

ANSWER_CHOICES:

[FREE_RESPONSE]

END_QUESTION

QUESTION_ID: 2c3aefc9

QUESTION:

In rectangle PQRS, the length of line segment QS is $\sqrt{2,344}$ and the length of side PS is 8 more than the length of side RS. What is the length of side QR?

ANSWER_CHOICES:

[FREE_RESPONSE]

END_QUESTION

QUESTION_ID: 2757549e

QUESTION:

In the figure, BC is parallel to DE. If the length of DE is 147 and the length of AE is 245, what is the value of tan x°?

ANSWER_CHOICES:

A. 147/196

B. 147/245

C. 196/147

D. 245/147

END_QUESTION

QUESTION_ID: c4a9c94d

QUESTION:

In the figure, AD and BC intersect at point D. If the length of BD is 5 and the tangent of angle CAD is 1.3, what is the length of AD?

ANSWER_CHOICES:

[FREE_RESPONSE]

END_QUESTION

QUESTION_ID: f389569d

QUESTION:

What is the perimeter of an equilateral triangle with a height of $39\sqrt{3}$?

ANSWER_CHOICES:

A. 117

B. $117\sqrt{3}$

C. 234

D. $1,521\sqrt{3}$

END_QUESTION

QUESTION_ID: e3158182

QUESTION:

Acute angle P has a measure of p radians, and acute angle T has a measure of t radians. The function g is defined by $g(x) = p(8)^x + 6t$. If $\sin p = \cos t$, which of the following represents the value $g(0)$ in terms of t?

ANSWER_CHOICES:

A. $6t + \pi$

B. $6t$

C. $5t + \pi$

D. $5t + \pi/2$

END_QUESTION

QUESTION_ID: 98e02472

QUESTION:

In right triangle XYZ, angles X and Z are acute angles. Point R lies on XZ such that YR is perpendicular to XZ. The measure of ∠XZY is a°, the measure of ∠XYR is b°, and the length of XY is 19. If sin a° = 12/13, what is the value of tan b°?

ANSWER_CHOICES:

[FREE_RESPONSE]

END_QUESTION

QUESTION_ID: de9148c4

QUESTION:

A right triangle has legs of length 12 and 3, and hypotenuse c. Which equation shows the relationship between the side lengths of the triangle shown?

ANSWER_CHOICES:

A. $12 \cdot 3 = c$

B. $12 + 3 = c$

C. $12^2 + 3^2 = c^2$

D. $12^2 - 3^2 = c^2$

END_QUESTION

QUESTION_ID: 4fb972f2

QUESTION:

In right triangle QRS, the length of hypotenuse SQ is $4\sqrt{82}$ and the length of side QR is 4. What is the length of side RS?

ANSWER_CHOICES:

[FREE_RESPONSE]

END_QUESTION

QUESTION_ID: 6c67dd47

QUESTION:

In the figure shown, the measure of angle X is 54°. The length of XY is 26 units and the length of XZ is 19 units. What is the area, in square units, of triangle XYZ?

ANSWER_CHOICES:

A. 247

B. 494

C. 247 sin 54°

D. 494 sin 54°

END_QUESTION

QUESTION_ID: e89f63bf

QUESTION:

The sum of the measures of two angles in a triangle is 64°. What is the measure of the third angle?

ANSWER_CHOICES:

A. 26°

B. 52°

C. 116°

D. 128°

END_QUESTION

QUESTION_ID: ecc98c87

QUESTION:

In the figure shown, line k intersects lines r and s. If w = 147, which additional piece of information is sufficient to prove that lines r and s are parallel?

ANSWER_CHOICES:

A. x = 33

B. y = 147

C. w + y = 180

D. y + z = 180

END_QUESTION

QUESTION_ID: 1dc7e423

QUESTION:

In triangle ABC, the measure of angle A is 52° and AC = 30. In triangle PQR, the measure of angle P is 52° and PR = 120. Which additional piece of information is sufficient to prove that triangle ABC is similar to triangle PQR?

ANSWER_CHOICES:

A. AB = 50 and PQ = 50.

B. AB = 50 and QR = 200.

C. The measures of angle B and angle R are 32° and 96°, respectively.

D. The measures of angle B and angle Q are 52° and 32°, respectively.

END_QUESTION

QUESTION_ID: b5d62bba

QUESTION:

In the figure, LQ intersects MP at point R, and LM is parallel to PQ. The lengths of MR and RP are 8 and 14 units, respectively. The area of triangle LMR is 36 square units. What is the area of triangle PQR, in square units?

ANSWER_CHOICES:

A. 576/49

B. 144/7

C. 63

D. 441/4

END_QUESTION

QUESTION_ID: 4a141e77

QUESTION:

In the figure, line p is parallel to line r, and line t intersects both lines. At the upper intersection, x° and 72° are vertical angles. What is the value of x?

ANSWER_CHOICES:

A. 36

B. 72

C. 180

D. 252

END_QUESTION

QUESTION_ID: 7a11ceea

QUESTION:

In the figure, segment AD and segment BE intersect at point C. Segment AD is perpendicular to line ℓ at point A. Which of the following additional pieces of information is(are) sufficient to determine that triangle ABC is congruent to triangle DEC?

I. Segment AD is perpendicular to line m at point D.

II. AC = 13 and DC = 13.

ANSWER_CHOICES:

A. I is sufficient by itself, but II is not.

B. II is sufficient by itself, but I is not.

C. I is sufficient by itself, and II is sufficient by itself.

D. Neither I nor II is sufficient by itself.

END_QUESTION

QUESTION_ID: 858809e7

QUESTION:

In the figure, EG and AD intersect at point F, and line ℓ is parallel to line j. If EF = 0.5CD and FG = 2.25CD, which of the following must be true?

I. AB = 2.25EF

II. FG = 0.5AB

III. EF/FG = AF/FD

ANSWER_CHOICES:

A. I only

B. II only

C. I and II

D. II and III

END_QUESTION

QUESTION_ID: d21270da

QUESTION:

In the figure, line t intersects lines r and s. One interior angle at line r is 24°, and the corresponding interior angle at line s is x°. Which additional piece of information is sufficient to prove that lines r and s are parallel?

ANSWER_CHOICES:

A. x < 24

B. x = 24

C. x > 24

D. x - 24 = 360

END_QUESTION

QUESTION_ID: 697b3954

QUESTION:

In the figure shown, triangle ABC is congruent to triangle BDE. If $r = 54$ and $s = 26$, what is the length of CD?

ANSWER_CHOICES:

A. 26

B. 27

C. 28

D. 54

END_QUESTION

QUESTION_ID: ffc39dc5

QUESTION:

In the figure shown, CD = EF and FC = DE. If $x = 65$ and $y = 30$, what is the measure of $\angle CDE$?

ANSWER_CHOICES:

A. 30°

B. 65°

C. 95°

D. 130°

END_QUESTION

QUESTION_ID: 8364487a

QUESTION:

Triangle TUV is dilated by a scale factor of 1/2 to obtain triangle T'U'V' (not shown), where T corresponds to T' and U corresponds to U'. In triangle TUV, angle V measures 25°. What is the measure, in degrees, of angle V'?

ANSWER_CHOICES:

A. 25

B. 38

C. 50

D. 76

END_QUESTION

QUESTION_ID: 0013adbd

QUESTION:

In the figure, line t intersects lines j and k. At the intersection with line j, an angle is 134° and the adjacent angle is w°. At the intersection with line k, the angles are labeled x°, y°, and z°. Which additional piece of information is sufficient to prove that lines j and k are parallel?

ANSWER_CHOICES:

A. x = 46

B. y = 46

C. w = 134

D. z = 134

END_QUESTION

QUESTION_ID: 1dacfb94

QUESTION:

In triangle ABC, AB = BC. Points D and E lie on line segments AB and BC, respectively, such that line segments DE and AC are parallel. Point O is the midpoint of line segment DE, and the measure of ∠OAC is 30°. Point F lies on line segment AC, and line segment BF passes through point O. If AB = 81 and AO = BO, what is the length of line segment BE?

ANSWER_CHOICES:

[FREE_RESPONSE]

END_QUESTION